Model Sensitivity
Supporting Document: `IndigoCarbon_US-1_2023_0072`

June 28, 2024

Data that drives DayCent-CR Version 1.1.0 and GGIT modeling systems in the Soil Metrics API

Relative Data Priorities

1st tier priority to enter correctly – these data have a major impact on the model results:

- Cash crop type per year/crop rotation
- Amount and N content of synthetic N fertilizer and/or organic amendments applied
- Tillage systems and year of transition to reduced tillage or no-till
- Presence or absence of cover crops and year of transition to using cover crops
- Crop residue removal (e.g., harvesting straw or corn stover)
- Burn events
- Location (Latitude and Longitude), daily weather information (i.e., maximum and minimum temperature and precipitation
- Soil textural class (% sand, silt, and clay contents)
- Presence of irrigation

2nd tier priority to enter correctly – these data have a moderate impact on the model results:

- Date of fertilizer events (+/- 3 days)
- Type of fertilizer (e.g., urea-based or not)
- Presence or absence of a hay or pasture in rotation in the current period (post-2000)
- Grazing utilization, start and end date, and rest period
- Irrigation date and amount

3rd tier priority to enter correctly – these data have a small impact on the model results:

- Presence or absence of a hay or pasture in the pre-2000 crop history
- Enrollment of Conservation Reserve Program in the pre-2000 crop history
- Date of harvest events (an approximation +/- one week)
- Date of plant events (an approximation +/- one week)
- Date of organic matter additions (manure/compost) (an approximation of +/- one week)

4th tier – data used only for QC/Debugging, or is a placeholder for future model improvements

- Yield

- N Application Method
- Liming applications

Summary of Changes

Section	Date	Summary of Changes
Relative Data Priorities	June 14, 2024	Date of fertilizer events and Type of fertilizer moved from 3rd tier to 2nd tier due to new model validation